

Chujia Guo

chujiaguo@gmail.com | 626-313-7468 | [linkedin.com/in/guo-chujia](https://www.linkedin.com/in/guo-chujia) | github.com/ChujiaGuo | chujia.dev

Education

University of Maryland, College Park

Master of Science in Data Science

College Park, MD

Sep 2026 – Present

University of Maryland, Baltimore County

Bachelor of Science in Computer Science

Baltimore, MD

Sep 2022 – May 2026

Technical Skills

Languages: Python, SQL

Frameworks: TensorFlow

Developer Tools: Git, Docker, AWS, Google Cloud Platform

Libraries: pandas, NumPy, Matplotlib

Experience

Guangzhou Quantitative Software Technology Co.

Jun 2025 – Aug 2025

Full-Stack Software Engineering Intern

Guangzhou, China

- Optimized Go backend APIs to broadcast real-time stock data to multiple concurrent internal clients via a proof-of-concept WebSocket implementation, reducing API response latency by 30%.
- Engineered a Next.js portfolio dashboard using split React Context windows to isolate high-frequency tick updates by ticker, reducing component re-renders by 25%.

Projects

Pavilion (Badminton Session & Skill-Rating Platform)

May 2026 – Present

Solo Developer

pavilion.chujia.dev

- Designed a weighted ranking algorithm that converges player ratings within 3–5 placement matches, incorporating volatility damping and time-decay factors validated across 25 unit and integration tests.
- Guaranteed zero double-booking anomalies and verified 100% of accessible database routes under concurrent load, authoring 61 automated tests and a live PostgreSQL testing container using pessimistic locking.
- Architected a domain-driven modular monolith using Next.js, Hono, and PostgreSQL to isolate 5 core domains for seamless microservices migration.

Classic (AI Student Scheduling Tool)

Oct 2025 – Jan 2026

Solo Developer

github.com/ChujiaGuo/classic

- Eliminated 100% of token costs and reduced parsing latency to sub-second speeds on duplicate uploads, architecting a multi-provider LLM abstraction layer with an SHA-256 and schema-versioned SQLite cache.
- Reduced average academic schedule generation time by over 90%, from 30 minutes of manual cross-referencing to under 2 minutes, exporting results as .ics files compatible with Google, Apple, and Outlook calendars.

BlueBot

Mar 2020 – May 2025

Founder & Lead Developer

github.com/ChujiaGuo/blueBot

- Automated player identity verification with >95% accuracy in under 10 seconds, integrating Google Cloud Vision and Tesseract OCR pipelines with Levenshtein distance matching across 10k+ player records.
- Engineered an asynchronous, event-driven bot architecture scaling to 10k+ members; implemented a hot-reload command system via runtime module cache-clearing and a process manager to block concurrent executions and prevent race conditions on shared state.